Statistical Mechanics: Recurrence, Nonequilibrium Order, and Ising Spins

Lecture by Leonard Susskind

Lecture notes organized by LazyingArt LLC with Video2Book

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Chapter 1

Recurrence, Nonequilibrium Order, and Ising Spins

Overview. A gas released from one half of a room rushes toward equilibrium, yet reversible mechanics says that sufficiently long evolution must also produce returns to the exceptional state. We estimate that return time from phase-space volume, separate recurrence from the unexplained low-entropy beginning of the universe, and use the resulting puzzle to discuss Boltzmann brains, coherent history, and life maintained by an energy flow. We then turn to magnetic order. Independent two-state spins give an exactly soluble warm-up: combinatorics determines the partition function and the magnetization is a hyperbolic tangent. Neighbor interactions finally restore an up–down symmetry and lead to the one-dimensional Ising model.

1.1 A Roomful of Air That Eventually Returns

Begin with all the air in the left half of a room. One can prepare this state with a partition: evacuate the right half, place the air on the left, and then remove the wall. The gas spreads rapidly through the room and approaches thermal equilibrium. Entropy increases. Nothing surprising has happened.

Now leave the room sealed and wait. A million consecutive heads is an extraordinarily unlikely coin-toss record, but an indefinitely repeated experiment eventually produces extraordinarily unlikely records. The same logic applies to the gas. At some remote time all of its molecules will again be found in the left half, or in some comparably special region. Such a return is a Poincare recurrence.

The precise theorem is slightly more careful than the picture. For a finite-measure region of phase space whose dynamics preserves phase-space volume, almost every initial point returns arbitrarily close to its starting point. A macroscopic statement such as *all molecules are in the left half* is easier: it asks only that the trajectory re-enter a finite coarse-grained region, not that every molecular coordinate return exactly. The practical question is how long that takes.

1.1.1 A high-dimensional phase point

For N classical particles, phase space has $6N$ coordinates,

$$\Gamma = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N; \boldsymbol{p}_1, \ldots, \boldsymbol{p}_N). \tag{1.1}$$

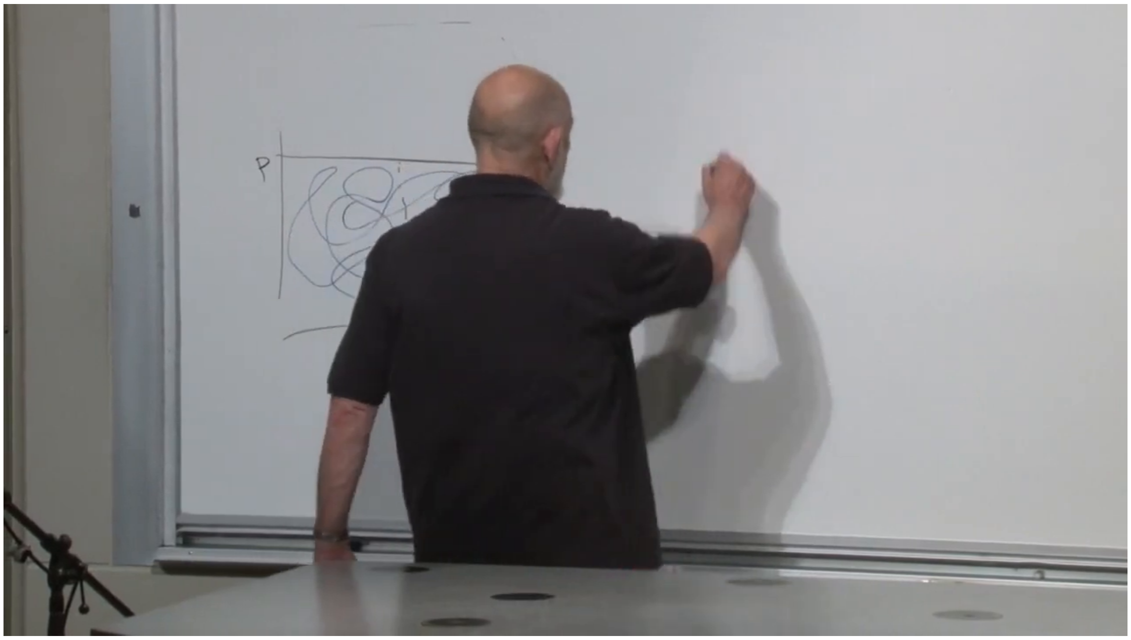

Figure 1.1: A schematic phase-space region and a complicated trajectory wandering through it. The drawing is deliberately low-dimensional; the real gas lives in a $6N$-dimensional space.

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The positions are bounded by the room. At fixed total energy the momenta are bounded as well, because one particle cannot carry arbitrarily large momentum without carrying arbitrarily large energy. The allowed energy shell therefore has finite measure under the assumptions of the recurrence argument.

The phase point moves deterministically, but in a many-body gas its motion is chaotic. Nearby trajectories separate quickly, as they do in a sequence of billiard-ball collisions. After coarse-graining, a long trajectory appears to wander throughout the accessible region.

The low-dimensional sketch immediately tempts us into a mistake. If the special state occupied half of the drawn rectangle, one might say that the gas should spend half its time there. But *every molecule lies in the left half* constrains every position coordinate, not just one coordinate of the phase point.

Take only two particles moving in one spatial dimension, with coordinates x_1 and x_2. Their configuration space is a square. Requiring $x_1 < L/2$ selects half the square; requiring both $x_1 < L/2$ and $x_2 < L/2$ selects one quarter:

$$P_2 = \frac{1}{2}\frac{1}{2} = \frac{1}{4}. \tag{1.2}$$

Three particles give $1/2^3$, and N particles give

$$P_{\text{half}}(N) = \left(\frac{1}{2}\right)^N. \tag{1.3}$$

The exponent is the entire point. A factor of one half is ordinary; one half raised to a macroscopic particle number is beyond ordinary intuition.

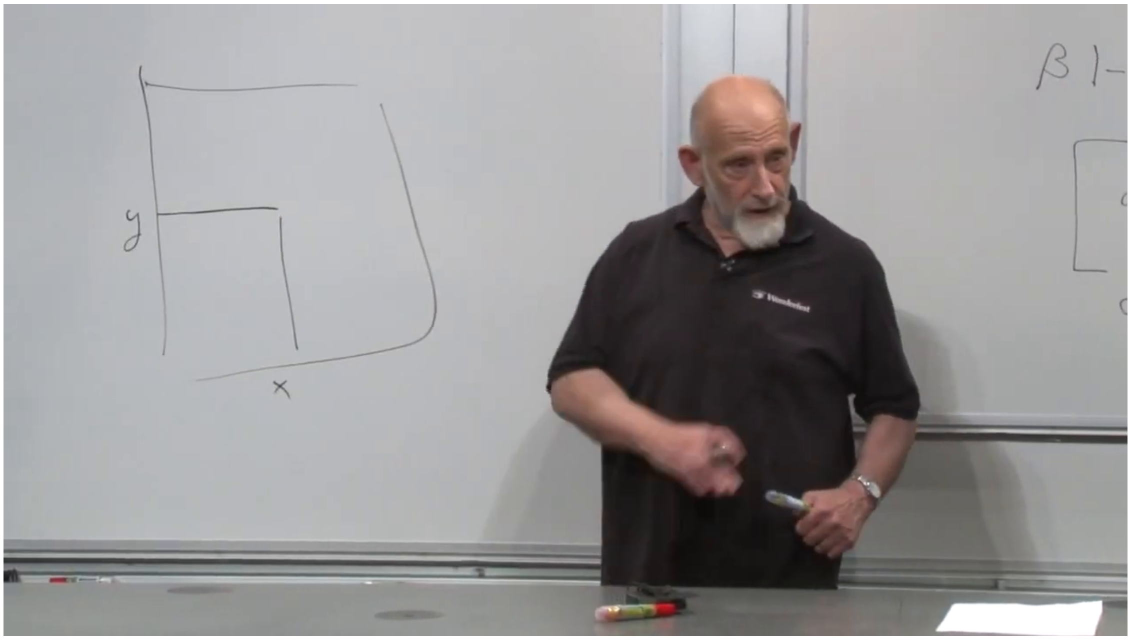

Figure 1.2: For two one-dimensional particles, the smaller square represents the configurations in which both coordinates lie in the chosen half of the room.

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> **Question from the audience.** Should the answer be $1/3^N$, since the molecules also have other spatial directions?
>
> **Response.** It depends on the condition being imposed. If the only question is whether every particle lies to the left of the dividing wall, then each particle satisfies one binary condition and the answer is 2^{-N}. Constraining another independent fraction of its available position space would supply another factor. [1]

1.2 Phase Volume, Entropy, and the Recurrence Scale

The same estimate can be written without referring to halves. Let the room have ordinary spatial volume V, and let v be a smaller region in which we demand that every molecule be found. Ignoring momentum factors that are common to the two macrostates, the available configuration-space volumes are

$$\Omega_{\text{room}} \propto V^N, \qquad \Omega_{\text{small}} \propto v^N. \tag{1.4}$$

The long-time fraction spent in the small macrostate is therefore

$$P(v) = \frac{\Omega_{\text{small}}}{\Omega_{\text{room}}} = \left(\frac{v}{V}\right)^N. \tag{1.5}$$

[1] Lecture timestamp: 00:07:47–00:08:05.

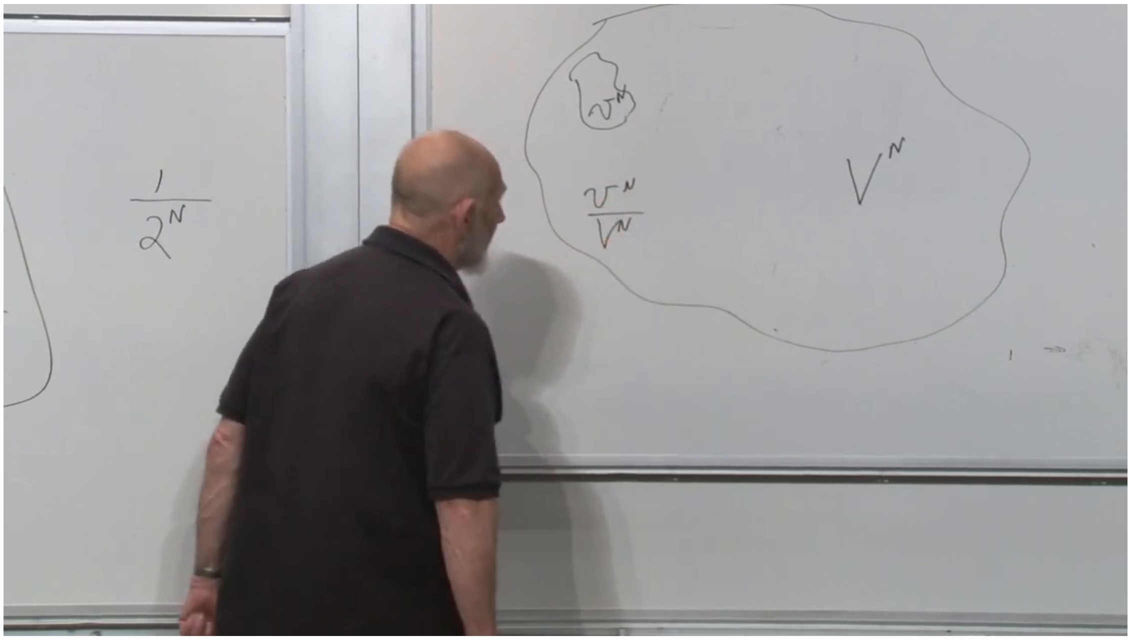

Figure 1.3: The board's general phase-volume picture: a small region v^N inside an accessible region V^N, with the ratio v^N/V^N.

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In units with $k_B = 1$, entropy is the logarithm of accessible phase volume. If $S_A = \log \Omega_A$ is the entropy of the constrained macrostate and $S_{\text{eq}} = \log \Omega_{\text{eq}}$ that of equilibrium, then

$$\begin{aligned} P(A) &= \frac{\Omega_A}{\Omega_{\text{eq}}} \\ &= \exp(S_A - S_{\text{eq}}) \\ &= e^{-\Delta S}, \\ \Delta S &= S_{\text{eq}} - S_A. \end{aligned} \tag{1.6}$$

For the half-room state, $\Delta S = N \log 2$, and (1.6) reduces exactly to 2^{-N}. The shorthand $P \sim e^{-S_{\text{eq}}}$ is appropriate only when the entropy of the special region is being treated as a fixed reference; the physically meaningful exponent is the entropy deficit ΔS.

> **Question from the audience.** Is this the volume of configuration space rather than the whole phase space?
>
> **Response.** For this comparison, yes. We are imposing a positional condition while leaving the momentum distribution at the same temperature, so the common momentum-space factor cancels. A fully general recurrence statement uses phase space; this particular probability can be computed from the configuration-space ratio. [2]

Suppose the system loses memory of its detailed position on a microscopic correlation or mixing time τ_{mix}. A rare region of probability $P(A)$ is then encountered on the scale

$$\tau_{\text{rec}} \sim \frac{\tau_{\text{mix}}}{P(A)} \sim \tau_{\text{mix}} e^{\Delta S}. \tag{1.7}$$

[2] Lecture timestamp: 00:11:21–00:11:40.

For the half-room condition,

$$\tau_{\text{rec}} \sim \tau_{\text{mix}} 2^N. \tag{1.8}$$

The lecture uses the deliberately rough room-sized estimate $N \sim 10^{30}$, giving the memorable scale $\tau_{\text{rec}}/\tau_{\text{mix}} \sim 2^{10^{30}}$. At ordinary atmospheric density a room is closer to 10^{27} molecules, but replacing 10^{30} by 10^{27} changes none of the conclusion: the recurrence time remains inconceivably larger than the age of the universe.

> **Question from the audience.** In what units is the recurrence time measured? Could a change from seconds to hours, or to an extremely short unit, alter the conclusion?
>
> **Response.** The dimensional prefactor is τ_{mix}; the huge number is the dimensionless ratio $\tau_{\text{rec}}/\tau_{\text{mix}}$. Changing seconds to hours divides by an ordinary conversion factor, which is negligible beside $2^{10^{30}}$. The duration of an individual visit to the corner does depend on molecular speeds. The estimate concerns how rarely the visit begins. There is also a quantum recurrence theorem, but it is not needed here. [3]

> **Question from the audience.** What is the objective of calculating a time that no experiment can possibly wait for?
>
> **Response.** The point is conceptual. It tells us in what sense a closed mechanical system remains reversible. It spends almost all its time near equilibrium, yet on sufficiently long scales it also makes fluctuations toward lower entropy. The second law is therefore an overwhelmingly probable statement about special initial data, not a microscopic equation that forbids reverse motion. [4]

1.3 Reversibility and the Special Beginning

Start knowingly in the tiny half-room region. Almost every neighboring trajectory immediately leaves it, because vastly more phase volume lies outside than inside. The gas spreads quickly and then waits an immense time before returning. This asymmetry between rapid relaxation and long waiting does not violate reversible mechanics.

The time reverse of the rapid escape is not the entire waiting period. It is the brief, fantastically well-coordinated episode in which equilibrium-looking molecular velocities happen to arrange themselves so that the gas *swooshes* into one half of the room. Reverse every momentum at the end of the ordinary escape and the gas retraces the escape in reverse. The long waiting time is the time required for that special incoming velocity pattern to occur spontaneously.

> **Question from the audience.** Does time symmetry mean that the slow recurrence into the corner takes as long as the original escape from it?
>
> **Response.** No. Compare the escape with its exact time reverse. Both are rapid. What takes an enormous time is waiting in equilibrium for the precise incoming fluctuation that initiates the reverse swoosh. Over an enormously long record, excursions away from equilibrium occur in both time directions with symmetric local histories. [5]

[3] Lecture timestamp: 00:17:24–00:19:18.
[4] Lecture timestamp: 00:15:12–00:16:35.
[5] Lecture timestamp: 00:20:46–00:21:58.

The recurrence argument consequently explains only part of the arrow of time. It explains why entropy ordinarily rises *given* that the system began in a low-entropy region. It does not explain why the initial condition was special. In cosmology that missing premise becomes the central question: why did the observable universe begin in such an exceptionally small region of its gravitational phase space?

> **Question from the audience.** Does recurrence apply to cosmology? After the universe spreads out, might it recur if one waits long enough?
>
> **Response.** That question is unavoidable, but its answer depends on global cosmology. A genuinely closed system with a finite equilibrium state space invites recurrence. An expanding spacetime, gravity, horizons, and an infinite or nonstationary state space can change the premises. The important lesson here is narrower: an equilibrium recurrence picture by itself is not a satisfactory statistical explanation of the structured universe we see. [6]

> **Question from the audience.** Does the recurrence depend on preparing the gas in a small region at the beginning? If the room starts at equilibrium, will the air still eventually gather into one corner?
>
> **Response.** Yes. Recurrence concerns the long-time dynamics of the closed system, so an equilibrium-looking initial state also undergoes rare low-entropy fluctuations. Preparing the half-room state matters for a different reason: it lets us condition on a known special beginning and ask why entropy rises immediately afterwards. [7]

1.4 Why a Random Fluctuation Predicts Too Little

One might try to turn recurrence into a cosmology: perhaps the universe spent almost all of eternity in equilibrium, and our present world is one rare fluctuation. The proposal fails a sharp statistical test. Condition on the existence of an observer and ask what else that observer should expect to see.

A fluctuation large enough to make one planet is extraordinarily rare. A fluctuation that makes two planets requires a larger entropy deficit and is still rarer. A fluctuation that makes stars, galaxies, a cosmic microwave background, and mutually consistent records of a long past is rarer by an enormous additional exponential. An equilibrium-fluctuation theory therefore predicts the smallest fluctuation sufficient for observation, not the richly coherent world actually observed. In its most extreme form, the minimal observer is called a Boltzmann brain.

The logic is conditional probability. Let A be the event that one observer-sized fluctuation occurs and B an independent second structure. If both have tiny probability p, then

$$P(B \mid A) = p, \qquad P(A \cap B) = p^2. \tag{1.9}$$

Learning that the first fluctuation occurred does not make the second likely. The lottery analogy is exact at this level: a previous winner has no improved chance on the next independent draw. Thus the observer may condition on being present, but should still be astonished to find an independently structured environment.

[6] Lecture timestamp: 00:19:59–00:20:45.
[7] Lecture timestamp: 00:22:04–00:22:20.

Question from the audience. Given that one planet or one Boltzmann observer already exists, is the probability of another one really smaller?

Response. The conditional probability for a second independent rare fluctuation is the same tiny one-event probability, not the square of it. The joint probability for both is the square. This is precisely why the first observer should still expect thermal disorder outside itself. Finding a spouse, another planet, and an ordered sky is evidence against the premise that the observer was assembled as an isolated equilibrium fluctuation. [8]

1.4.1 Coherent records of a past

The same argument applies to history. Two textbooks that agree, astronomical records that fit one expansion history, geological strata, memories, and laboratory notebooks all constrain one another. A single random textbook could contain any story. An entire network of records that agrees about the same past occupies a much smaller phase-space region.

The classroom example uses the familiar tale that George Washington chopped down a cherry tree. Whether that tale is true is irrelevant. The point is that, in a world fluctuated into existence at the present instant, one book's sentence gives no dynamical reason for another record to agree with it. Coherent correlations are evidence of a shared history, not merely of one instantaneous accident.

Question from the audience. What does *historical coherence* mean here?

Response. It means that many present records fit one another as consequences of a common past. A fluctuation could fabricate records and memories, but fabricating a mutually consistent network requires a much larger fluctuation than fabricating one minimal observer. The recurrence theory therefore predicts that most observer-like fluctuations lack the coherence we actually find. [9]

Question from the audience. Could the universe still be a closed system whose equilibration and recurrence times are merely extremely long?

Response. That possibility does not remove the statistical problem. Over an indefinitely long history there would be many observer-like fluctuations, and the overwhelming majority would be minimal ones without a coherent surrounding history. An observer selected from that ensemble should not expect our observations. Cosmologists therefore take the Boltzmann-brain problem seriously as a test of cosmological measures and late-time models, not as a claim that recurrence has already been observed. [10]

[8] Lecture timestamp: 00:26:12–00:30:27.
[9] Lecture timestamp: 00:30:30–00:31:12.
[10] Lecture timestamp: 00:31:13–00:32:42.

1.5 Life as Structure Sustained by Flow

Life seems to pose the opposite difficulty. Organisms maintain intricate organization for long times; they do not look like brief accidental fluctuations. The resolution is that the second law applies to a complete closed system. A subsystem may lower or maintain its entropy provided a larger entropy increase occurs in its environment:

$$\dot{S}_{\text{total}} = \dot{S}_{\text{subsystem}} + \dot{S}_{\text{environment}} \geq 0. \tag{1.10}$$

The Earth is not in thermal equilibrium. It is approximately stationary on human timescales, but stationary does not mean static. Energy flows through it. Sunlight arrives, is degraded through many processes, and is radiated away at a lower effective temperature. That throughput can sustain local organization while the total entropy rises.

Water flowing through a pipe supplies the immediate mechanical image. The flow may contain persistent eddies and vortices. They are organized structures, but they are maintained by the stream. Seal both ends, stop the flow, and viscosity erases the eddies as the water approaches a dull equilibrium. The outgoing water is warmer than the incoming water, so the larger system has paid the entropy bill.

> **Question from the audience.** Does life decrease entropy, and why can its organization persist rather than appear and disappear as an ordinary fluctuation?
>
> **Response.** Life may maintain or lower entropy locally, but only by exporting still more entropy. Its persistence comes from a stationary nonequilibrium flow, chiefly the flow of solar energy through the Earth. If the Earth were sealed against incoming and outgoing radiation, it would eventually approach thermal equilibrium and the organized activity would cease. Life is closer to an eddy in a stream than to an equilibrium fluctuation. [11]

With that, we leave the *interesting things* and return, in the lecture's deliberately dry phrase, to something dull: magnets.

1.6 Cooling a Magnet: Domains and a Chosen Direction

Statistical mechanics often calls a mathematical model a magnet even when it is not a literal piece of iron. The useful common feature is a collection of local degrees of freedom that can order. In a real magnetic material these degrees of freedom may arise from atomic magnetic moments, electron spins, orbital currents, or effective moments of crystal units.

At high temperature thermal disorder makes the local moments point in many directions, giving little or no net magnetization. If neighboring moments have lower energy when aligned, cooling allows small aligned domains to form. Further cooling makes the energy advantage more important and the correlated domains grow. In a system that supports a ferromagnetic transition, the correlation length becomes macroscopic at a finite critical temperature. The ordered phase then chooses an orientation even though the zero-field energy did not prefer one orientation in advance.

[11] Lecture timestamp: 00:32:43–00:35:42.

Question from the audience. Should microscopic magnets prefer to anti-align because a north pole attracts a south pole?

Response. The effective interaction depends on the material. Ordinary dipole intuition is not the whole microscopic problem; exchange, crystal structure, and competing interactions matter. Ferromagnetic materials are those in which the effective low-energy tendency favors parallel order, while other materials can favor anti-alignment. [12]

Question from the audience. Must an external magnetic field be present while the material cools in order for the moments to line up?

Response. No. The interaction can produce order without a field. In an exactly symmetric idealization, different directions are energetically equivalent, so the realized direction is selected by initial fluctuations, boundaries, or an arbitrarily small stray field. Choosing one of those equivalent directions is spontaneous symmetry breaking; the onset of macroscopic order is the magnetic phase transition. [13]

Real materials are complicated, so we now simplify brutally. Every local moment may point only up or down:

$$\sigma_i = +1 \quad \text{or} \quad \sigma_i = -1. \tag{1.11}$$

There is no east or west in this model. The two allowed orientations are a definition of the model, as heads and tails are the two outcomes of a coin. Its value is not microscopic realism but mathematical clarity and its ability to represent ordering phenomena shared by many systems.

Question from the audience. What are these two-state objects physically? Should they be imagined as particles?

Response. One may picture atoms on the sites of a crystal, each with an effective magnetic moment produced by spin or microscopic current. For the model, however, only the two-valued variable matters. The restriction to up and down is imposed to make the statistical mechanics simple. [14]

1.7 Independent Spins in an External Field

The first model contains N spins that do not interact with one another. An external field H supplies the only energy scale, and μ measures the strength of one spin's coupling to it. We keep the sign convention used in the calculation:

$$\epsilon_i = \mu H \sigma_i. \tag{1.12}$$

For $H > 0$, an up spin $\sigma_i = +1$ has energy $+\mu H$, while a down spin $\sigma_i = -1$ has energy $-\mu H$. Thus this convention makes the positive field favor down spins. It is the reverse of another common naming convention, but no physics depends on the names if the signs are used consistently.

Let n be the number of up spins and m the number of down spins. Then

$$n + m = N, \qquad E = (n - m)\mu H. \tag{1.13}$$

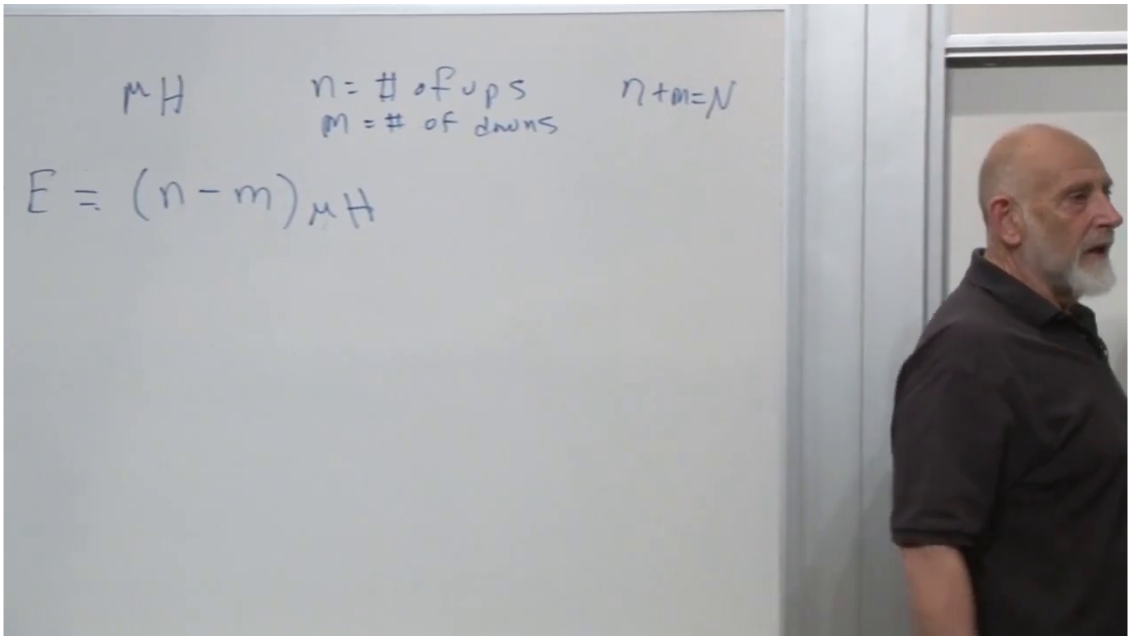

Figure 1.4: The independent-spin setup: n up spins, m down spins, $n+m=N$, and $E=(n-m)\mu H$.
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Question from the audience. What are the dimensions of H, μ, and E? Is H a number or a magnetic field?

Response. H denotes the externally imposed field strength and μ the magnetic moment. Their product μH has dimensions of energy. In the abstract model one often combines them into a single energy-valued parameter, customarily also denoted by a lowercase h. Here we retain both factors to show their physical origin, and E is the total energy. [15]

1.7.1 Counting configurations

For fixed n and $m = N - n$, choose which n of the N sites point up. The multiplicity is

$$\Omega(n,m) = \binom{N}{n} = \frac{N!}{n!\,m!}. \tag{1.14}$$

Every one of these configurations has the same energy (1.13).

Question from the audience. Can n and m both be summation indices, since they are not independent? And where does the combinatorial factor enter?

Response. They may be written as a constrained pair with $n+m=N$, or one may sum over $n = 0,\ldots,N$ and set $m = N-n$. The Boltzmann weight belongs to each individual configuration; grouping all configurations of the same energy multiplies that weight by $N!/(n!\,m!)$. [16]

[12] Lecture timestamp: 00:39:56–00:40:30.
[13] Lecture timestamp: 00:40:31–00:42:33.
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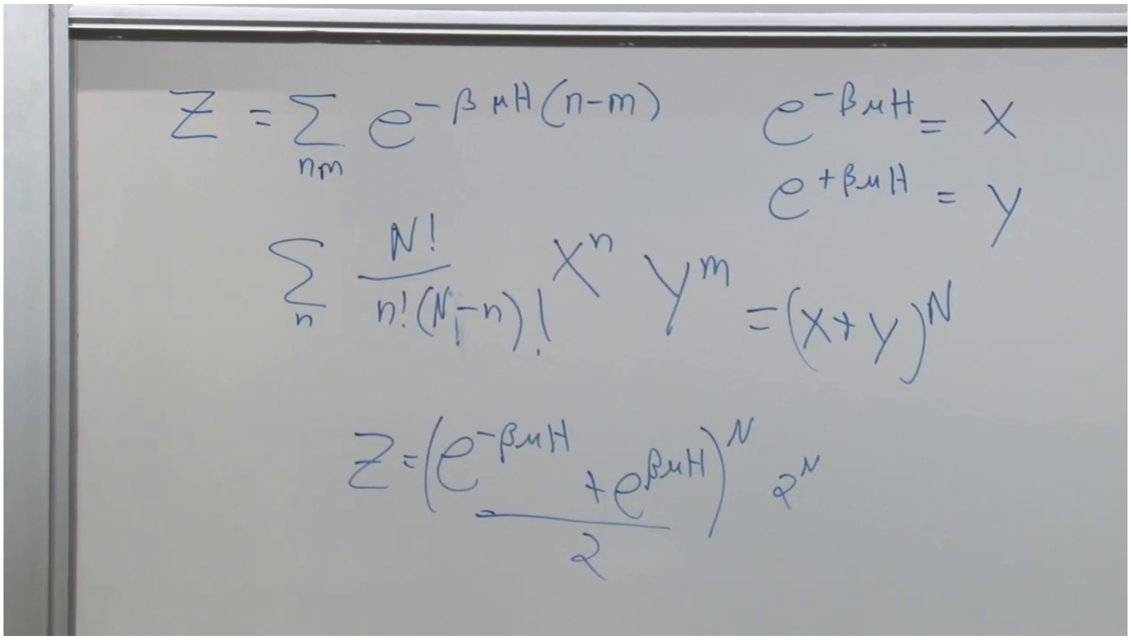

Figure 1.5: The partition-function sum, the substitutions x and y, and the binomial reduction to a power of $2\cosh(\beta\mu H)$.

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The partition function is therefore

$$Z_N = \sum_{n+m=N} \frac{N!}{n!\,m!} e^{-\beta\mu H(n-m)}$$
$$= \sum_{n=0}^{N} \binom{N}{n} \left(e^{-\beta\mu H}\right)^n \left(e^{+\beta\mu H}\right)^{N-n}. \tag{1.15}$$

Introduce

$$x = e^{-\beta\mu H}, \qquad y = e^{+\beta\mu H}. \tag{1.16}$$

The binomial theorem then completes the calculation:

$$Z_N = \sum_{n=0}^{N} \binom{N}{n} x^n y^{N-n} = (x+y)^N$$
$$= \left(e^{-\beta\mu H} + e^{+\beta\mu H}\right)^N = \left[2\cosh(\beta\mu H)\right]^N. \tag{1.17}$$

There is also a shorter derivation. Since the spins are independent, one spin has

$$z_1 = e^{-\beta\mu H} + e^{+\beta\mu H} = 2\cosh(\beta\mu H), \tag{1.18}$$

and $Z_N = z_1^N$. The combinatorial derivation is nevertheless worth keeping because it shows explicitly how multiplicity competes with energy.

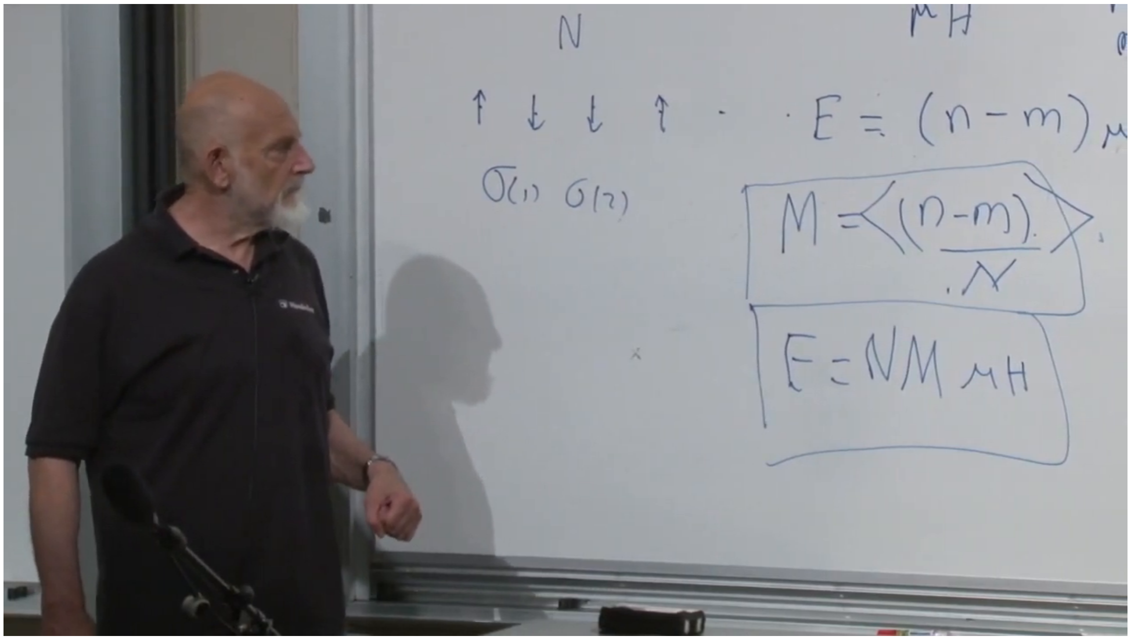

Figure 1.6: The spin row, the corrected average magnetization, and the energy relation emphasized on the board.

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1.8 Configuration Magnetization and Ensemble Magnetization

For one configuration define its dimensionless spin excess per site by

$$M_{\text{conf}} = \frac{n-m}{N}. \tag{1.19}$$

It ranges from $+1$ for all spins up to -1 for all spins down. The thermodynamic magnetization used here is its Boltzmann average,

$$M = \langle M_{\text{conf}} \rangle = \left\langle \frac{n-m}{N} \right\rangle. \tag{1.20}$$

The factor μH does not belong in this definition. It was briefly written there during the calculation and then removed; μH converts spin excess into energy.

Configuration by configuration,

$$E = N\mu H M_{\text{conf}}. \tag{1.21}$$

Linearity of averaging immediately gives

$$\langle E \rangle = N\mu H M. \tag{1.22}$$

Question from the audience. Is magnetization being assigned to a particular configuration, or are we discussing an average magnetization? Why does the same relation hold for average energy?

Response. Both notions appear, and they should be distinguished. $(n-m)/N$ is the value in one configuration; M is its Boltzmann average. Since $E = N\mu H M_{\text{conf}}$ holds separately in every configuration, averaging both sides with the same probabilities gives $\langle E \rangle = N\mu HM$. The fixed quantities N, μ, and H pass through the average. [17]

Question from the audience. How can the model align every spin north–south without biasing up against down?

Response. The absence of east and west is built into the two-state model; it is a restriction of the allowed degrees of freedom. The present external-field model *is* biased between up and down: with our sign convention the field lowers the energy of down spins. Consequently this model has no up–down symmetry. The zero-field Ising model introduced below will restore that symmetry. [18]

1.8.1 Energy from the partition function

The mean energy follows from the usual identity

$$\langle E \rangle = -\frac{\partial}{\partial \beta} \log Z_N. \tag{1.23}$$

Using (1.17),

$$\begin{aligned} \log Z_N &= N \log 2 + N \log[\cosh(\beta\mu H)]\,, \\ \langle E \rangle &= -N\mu H \, \frac{\sinh(\beta\mu H)}{\cosh(\beta\mu H)} \\ &= -N\mu H \tanh(\beta\mu H). \end{aligned} \tag{1.24}$$

Combining this with (1.22) gives

$$\boxed{M = -\tanh(\beta\mu H)}. \tag{1.25}$$

The shape of the answer contains the physics. Recall

$$\cosh x = \frac{e^x + e^{-x}}{2}, \qquad \sinh x = \frac{e^x - e^{-x}}{2}. \tag{1.26}$$

Near the origin, $\tanh x = x + O(x^3)$; for large positive x, it approaches 1. Since $\beta = 1/T$,

$$M \longrightarrow -1 \quad (T \longrightarrow 0), \qquad M \longrightarrow 0 \quad (T \longrightarrow \infty). \tag{1.27}$$

At zero temperature every spin occupies the lower-energy down state. At infinite temperature all configurations become equally weighted and the field bias is washed out. Reversing H reverses M.

Question from the audience. The magnetization changes continuously. Where is the phase-transition temperature that was mentioned earlier?

Response. There is none in this model. Independent spins in a nonzero external field are too simple, and $\log Z_N = N \log[2\cosh(\beta\mu H)]$ is smooth at every finite positive temperature. A phase transition requires the interacting, symmetric problem that we now approach; the two-dimensional zero-field Ising model is the first member of this family with a finite-temperature transition. [19]

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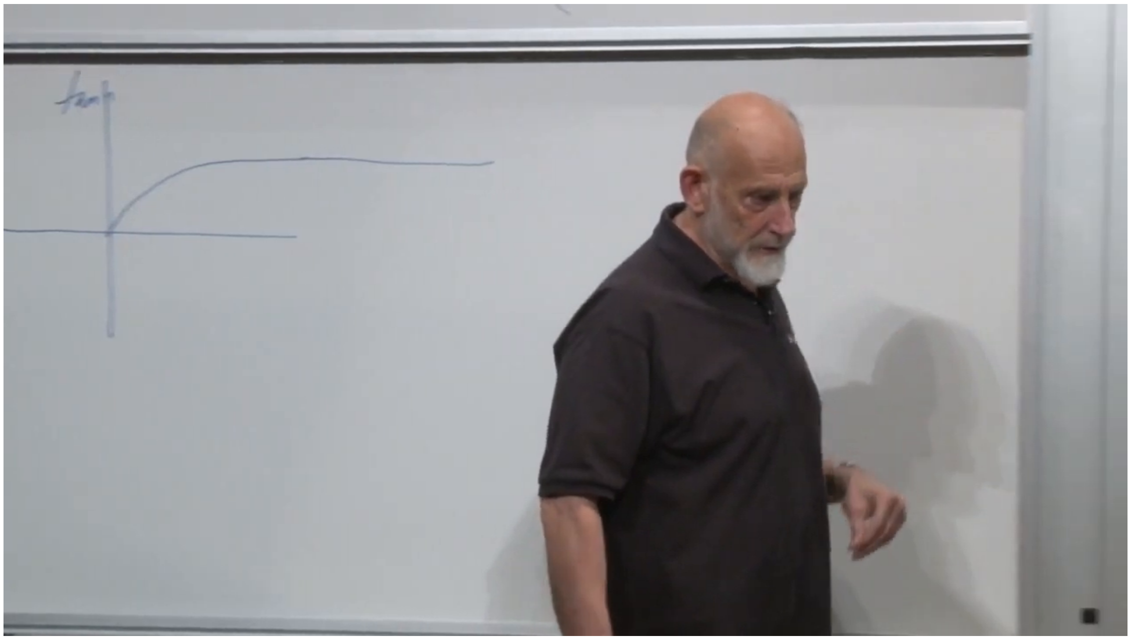

Figure 1.7: The hyperbolic-tangent curve approaches a constant at large $\beta\mu H$ and is linear near the origin. The physical magnetization in the lecture's convention is the negative of the plotted tanh.

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1.9 The Ising Model: Energy in Neighboring Pairs

The external field chose a direction before the statistics began. Remove that field and let neighboring spins provide one another's effective field. If both neighbors of a site point up, they favor up; if both point down, they favor down; if one points each way, their effects cancel in the simplest picture. Energy is now associated with pairs rather than isolated sites.

For two neighboring spins, choose $J > 0$ and write

$$E_{12} = -J\sigma_1\sigma_2. \tag{1.28}$$

Aligned spins have $\sigma_1\sigma_2 = +1$ and energy $-J$; anti-aligned spins have product -1 and energy $+J$. The minus sign is therefore exactly what makes the interaction ferromagnetic.

On an open one-dimensional chain the Hamiltonian is

$$\mathcal{H} = -J\sum_{i=1}^{N-1}\sigma_i\sigma_{i+1}. \tag{1.29}$$

With periodic boundary conditions one sets $\sigma_{N+1} = \sigma_1$ and sums through N. Either convention counts every neighboring pair once; their boundary difference becomes negligible for a long chain.

At infinite temperature every spin string is equally probable. Half the bonds are aligned on average and half anti-aligned, so

$$\langle\sigma_i\sigma_{i+1}\rangle_{T=\infty} = 0, \qquad \langle\mathcal{H}\rangle_{T=\infty} = 0. \tag{1.30}$$

Zero energy here is not a low-energy state. The all-aligned ground state has energy $-J$ per bond, so the infinite-temperature ensemble lies an extensive energy above it.

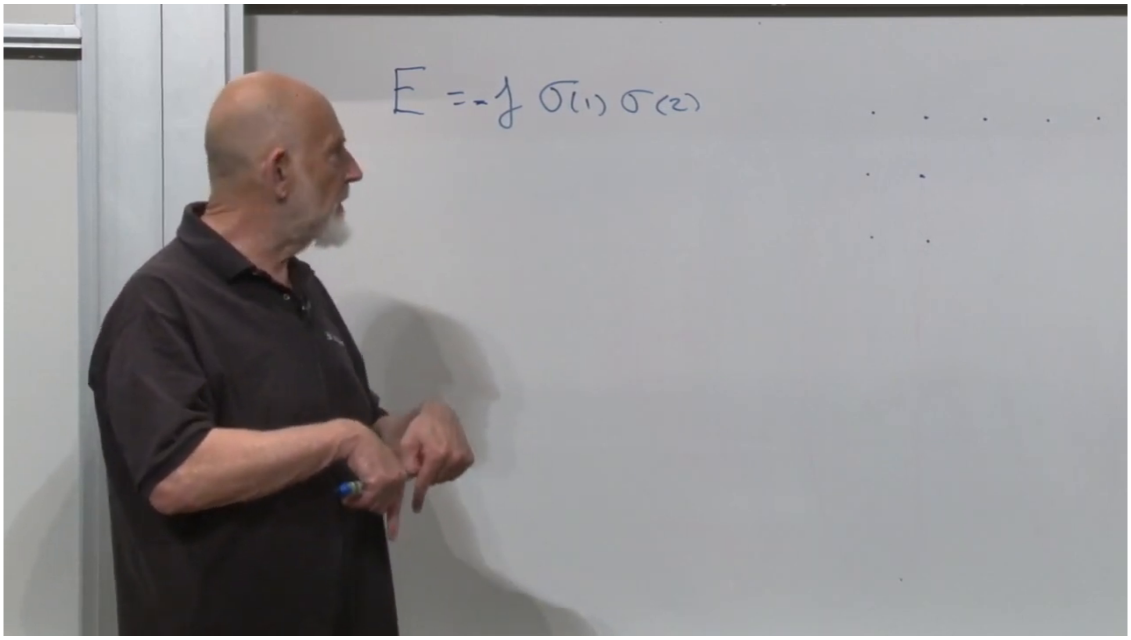

Figure 1.8: The local ferromagnetic interaction $E_{12} = -J\sigma_1\sigma_2$.
Lecture frame: 01:19:59.

Question from the audience. How can the average product be zero when every individual product $\sigma_i\sigma_{i+1}$ is either $+1$ or -1?

Response. An average need not be one of the values taken in a single configuration. At infinite temperature aligned and anti-aligned bonds occur equally often, so their $+1$ and -1 contributions cancel. The same averaging distinction that mattered for magnetization matters here. [20]

1.9.1 Two ground states and one global symmetry

At zero temperature every bond wants to be aligned. There are consequently two ground states,

$$(+, +, +, \dots, +) \qquad \text{and} \qquad (-, -, -, \dots, -). \tag{1.31}$$

They have exactly the same energy. The Hamiltonian makes this degeneracy transparent: reverse every spin,

$$\sigma_i \longmapsto -\sigma_i \qquad \text{for all } i, \tag{1.32}$$

and every bond product is unchanged,

$$(-\sigma_i)(-\sigma_{i+1}) = \sigma_i\sigma_{i+1}. \tag{1.33}$$

Flipping one spin is not a symmetry, because it reverses the sign of the adjacent bond products. Flipping all spins is a symmetry. This is the global $\mathbb{Z}_2$ symmetry of the zero-field Ising model.

Apply a tiny field to one spin at strictly zero temperature. It breaks the tie between the two ground states, and the ferromagnetic couplings propagate that choice through the chain. Remove the field

[20] Lecture timestamp: 01:21:39–01:22:37.

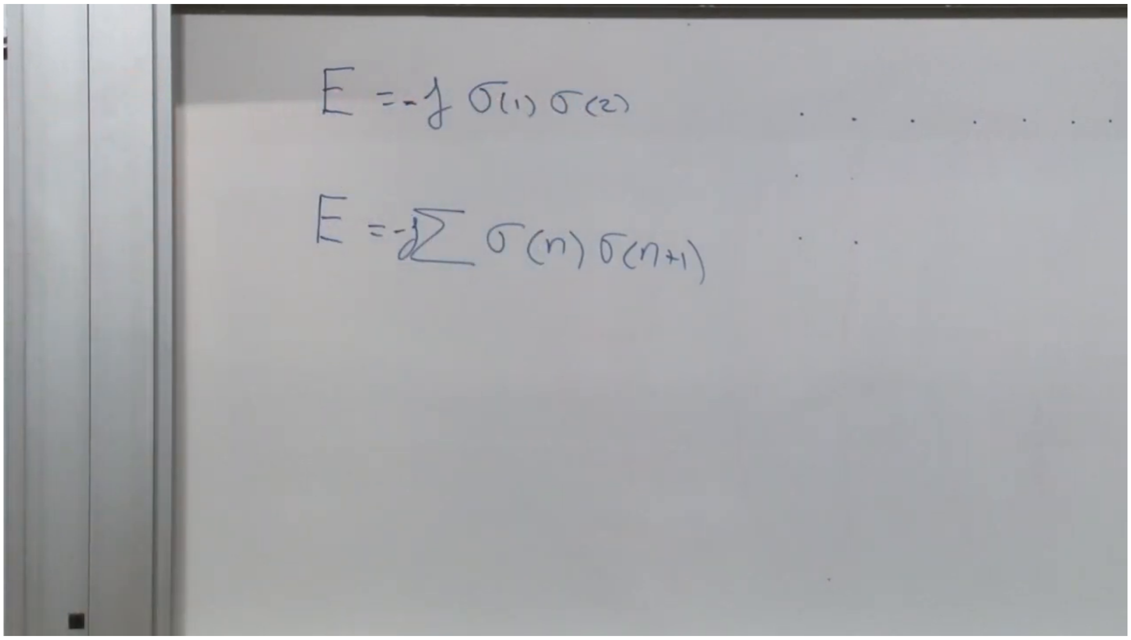

Figure 1.9: The local pair energy extended to the nearest-neighbor chain Hamiltonian.
Lecture frame: 01:20:46.

after the state has been selected and a large system can retain its orientation for a very long time. This is the intuitive entry point to spontaneous symmetry breaking.

There is one essential statistical qualification. For a finite system in exactly zero field, the canonical ensemble gives equal weight to the two ground states and hence zero average magnetization. A broken-symmetry phase is defined by taking the thermodynamic limit before removing the selecting field:

$$M_* = \lim_{H\to 0^+} \lim_{N\to\infty} M_N(H). \tag{1.34}$$

In the one-dimensional nearest-neighbor model this limit is nonzero only at $T = 0$; any finite temperature destroys long-range order in the infinite chain. In two dimensions the ordered phase survives below a nonzero critical temperature.

1.9.2 A historical correction

The classroom account gives this history with a joke at Ising's expense. The accurate chronology is kinder. Wilhelm Lenz proposed the model, and his student Ernst Ising solved the one-dimensional case in his 1924 thesis, published in 1925. Ising correctly found no finite-temperature ferromagnetic transition in one dimension, although the broader significance of higher dimensions was not yet understood. Kramers and Wannier developed the two-dimensional duality and identified the square-lattice critical point in 1941; Onsager gave the celebrated exact zero-field solution in 1944.

The next calculation is now clear. We must find the partition function of (1.29) and determine how its correlations depend on temperature. That calculation will show explicitly why the one-dimensional chain has no finite-temperature phase transition and why dimensionality matters.

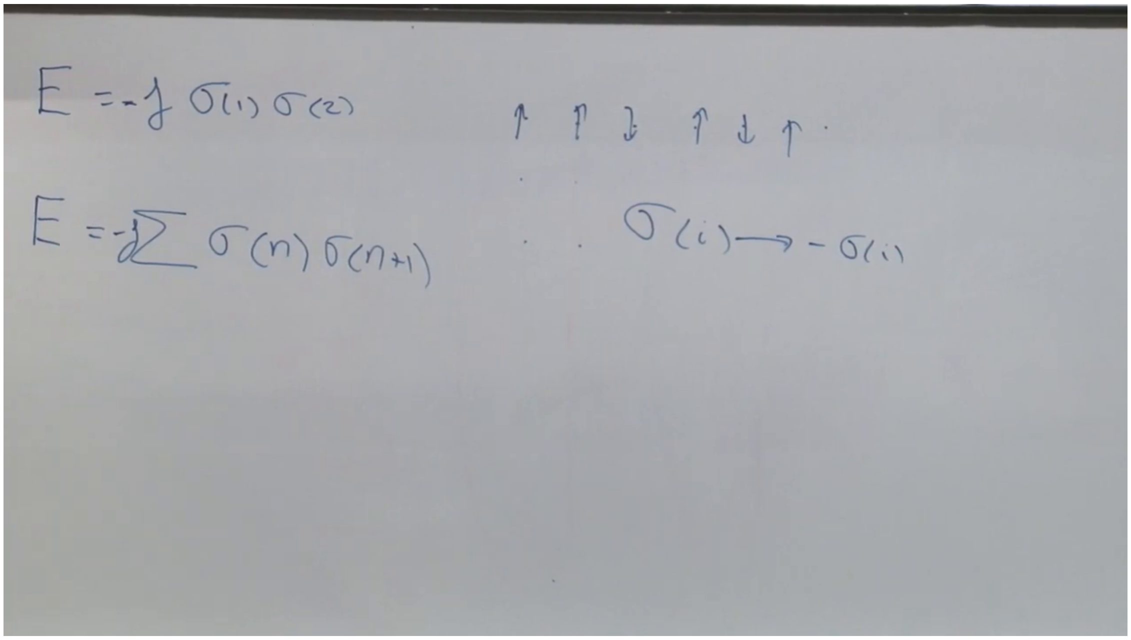

Figure 1.10: The chain Hamiltonian beside the global transformation $\sigma_i \mapsto -\sigma_i$, which leaves every pair product invariant.

Lecture frame: 01:26:31.

1.10 Where We Stand

The recurrence problem and the spin problem are two versions of the same statistical lesson. In the gas, an entropy deficit makes a macrostate exponentially rare:

$$P(A) = e^{-\Delta S}, \qquad \tau_{\text{rec}} \sim \tau_{\text{mix}} e^{\Delta S}. \tag{1.35}$$

Reversible mechanics permits the return, while the enormous phase-volume ratio explains why ordinary experience sees only relaxation. The remaining arrow-of-time problem lies in the special cosmological initial condition, not in a failure of microscopic reversibility.

For independent spins in a field,

$$Z_N = [2\cosh(\beta\mu H)]^N, \qquad M = -\tanh(\beta\mu H). \tag{1.36}$$

That system is explicitly biased and changes smoothly with temperature. The Ising interaction

$$\mathcal{H} = -J \sum_i \sigma_i \sigma_{i+1} \tag{1.37}$$

removes the external bias, restores a global up–down symmetry, and presents the real ordering question: can a symmetric many-body system choose one of two ordered states at nonzero temperature?